

TCS Labs' Adoption of Layer5 Meshery and Kanvas

TCS Labs runs Meshery as the design and collaboration plane of its Internal Developer Platform — multi-tenant, multi-identity-provider, and white-labeled for client engagements.

TCS Labs — the London Engineering Centre of Excellence within Tata Consulting Services (tata-consulting.co.uk), operating independently of core corporate IT governance — runs Meshery as the collaboration and control plane at the heart of a service provider-grade Internal Developer Platform. Layer5 hosts TCS Labs' multi-tenant, multi-identity-provider Meshery deployment: one management plane that maps a deep, multi-tiered enterprise hierarchy onto isolated tenants, each with its own identity provider, policies, and audit trail.

In production as the platform's design and collaboration surface since January 2024, Meshery serves 22 teams — 19 of them customer-facing teams designing directly with clients through a white-labeled Meshery.

Fast facts

22 teams in production. Meshery has served as the platform's design and collaboration surface since January 2024: 19 customer-facing teams design with clients via white-labeled Meshery; 3 internal teams publish custom models to a private OCI registry.	Multi-tenant, multi-IdP. One Layer5-hosted deployment maps a deep enterprise hierarchy onto isolated tenants, each with its own identity provider, policies, and audit trail.
Design plane, not deploy path. Designs export to per-organization private GitLab repos and deploy through Argo CD — updating Meshery never touches running client systems.	Open by default. Active upstream contributors to Meshery core and Meshery Operator, plus a Meshery Extensions maintainer grown through the contributor-to-maintainer pipeline.

The problem: a platform for hundreds of thousands of engineers

Most internal developer platforms are built for a single product team, then stretched painfully across the business. TCS Labs started at the other end: hundreds of thousands of engineers, deep hierarchies (group, business unit, account, sub-account, project, team, environment), and clients who care deeply about audit, residency, and isolation. Three properties were non-negotiable: complex organizational hierarchies mirrored in the platform's logical boundaries; strict multi-tenancy as the default; and policy-as-code RBAC with audit trails an auditor will actually read.

The studio's published numbers: 40+ UK consulting services engineers, 14 FTSE 100 clients on UK platforms, 400+ microservices managed across UK estates, and 99.97% platform availability across 2025.

Where Meshery sits in the stack

Meshery — extended in-house with custom operators, organization-aware tenancy, and policy enforcement — is the shared system of record for platform architecture and application topology: the surface where architects, platform engineers, security teams, and developers model, govern, and refine cloud-native components together. The team runs the Meshery v1.0 GA line on the stable channel, production pinned to specific stable releases with security fixes fast-tracked.

Why Meshery won

- **Collaboration as a first-class concern** — decisive for a platform whose whole point is shared, paved-road delivery.
- **Highly customizable, white-label UI** — customer-facing teams present Meshery as their own branded design surface.
- **Higher-level abstraction** over cloud-native YAML and JSON — iterate without drowning in manifests.
- **No lock-in** — designs export in multiple formats.
- **Model-driven breadth** — models incorporate Crossplane today, KRO next; choosing Meshery never meant giving those up.
- **Open core, CNCF-native** — contributable, in Go and TypeScript.

“Meshery is the only tool I know that treats collaboration — bringing engineers and teams together inside an engineering platform — as central to productivity rather than an afterthought. For a platform whose whole point is shared, paved-road delivery, that was decisive.”

— Mohd Hamza Shaikh, Platform Engineer and Tech Lead, TCS Labs

The Layer5 engagement: hosted multi-tenant, multi-IdP Meshery

TCS Labs pays Layer5 to host its Meshery deployment. Tenant boundaries follow the organizational hierarchy; isolation of compute, network, and data is the default. Because the deployment is Layer5-hosted, TCS Labs' platform team spends its time extending Meshery rather than operating the control plane. The relationship is visible in the open: a stock Meshery install ships with a built-in remote provider entry for the deployment at platform.tata-consulting.co.uk, and TCS Labs maintains its own Meshery remote-provider implementation exposing organizations, workspaces, and environments, with production guidance to federate each tenant's upstream identity provider via OIDC.

What changed

- **Faster ramp into cloud-native** — engineers and clients iterate on configurations without YAML overload.
- **Collaboration as the productivity unlock** — teams and clients work on the same designs.
- **Client engagement through a branded surface** — the white-labeled design tool.
- **Reuse across teams** — shareable designs and custom models in a private OCI registry.
- **No lock-in, clean handoff to GitOps** — export in multiple formats, deploy through Argo CD.
- **Multi-tenant, multi-IdP by design** — OIDC federation per tenant, SPIFFE/SPIRE workload identity, policy-as-code RBAC.

Measured impact, as published by TCS Labs: new-engagement environment readiness from 14 days to under a day; ~40% lower run cost versus bespoke per-account stacks; 12-week median to stand up a client tenant.

The people behind it

Platform engineer Mohd Hamza Shaikh entered the cloud-native world through a Meshery LFX mentorship in 2023, brought Meshery into TCS Labs when the team formed, and remains an active contributor to Meshery core and Meshery Operator — work that earned him GitHub's Pull Shark and Pair Extraordinaire badges. His engineer Arjun Mehta is now a Meshery Extensions maintainer: the project's genuine contributor-to-maintainer pipeline at work. “The warmth of the community,” Hamza says, “is exactly why I am still here.”

Open by default

TCS Labs' edge, in their own telling, “does not come from secrecy. It comes from being one of the most active enterprise contributors to the cloud-native projects underneath” — Meshery core, Meshery Operator, the lifecycle operators running the collaboration suite, and open tooling like helm-kanvas-snapshot, a Helm-aware utility for capturing and visualizing release state.

“The job of a platform team is to make the right thing the easy thing. Everything else — the diagrams, the Helm charts, the bank-grade controls — is in service of that single test.”

— Eleanor Whitcombe, Director, London Engineering Centre of Excellence

Honest gaps

Two gaps top Hamza's wish list: Meshery's marketing and visibility — “an enormous feature set but only a drizzle of blog posts” — and first-class Terraform support, a real gap given how much TCS Labs and its clients rely on Terraform. Also hoped for: deeper model coverage and a long-term-support channel for risk-averse adopters.

TCS Labs Academy

The partnership extends to education. TCS Labs Academy — on the shared Layer5 Academy platform — spans six learning paths and 23 courses (flagship: AI-Assisted Infrastructure Design with Meshery & Kanvas), five hands-on challenges, four importable Meshery designs, and a two-tier AI certification: CAINA (50 questions, 75 minutes, 70% to pass, plus a hands-on lab) and CAINP (65 questions, 90 minutes, 70% to pass, plus a mandatory capstone).

Overall, Meshery gives TCS Labs a single, open control plane for designing, governing, and collaborating on cloud-native infrastructure across a multi-tenant, multi-identity-provider estate — inside Tata and across client engagements. By hosting that plane with Layer5 and building on it in the open, TCS Labs turned an internal platform into a product: one paved road, hundreds of thousands of engineers, and clients who buy the story at the board level.